

KANSAS CITY WHEELER, MO, USA

WMO: 724463

Lat: 39.121N

Lon: 94.597W

Elev: 226

StdP: 98.64

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13988

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.6	-12.1	-20.7	0.6	-12.7	-18.5	0.8	-10.9	10.8	6.6	9.4	4.1	3.4	320	0.509

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	36.2	24.7	34.5	24.4	33.0	24.1	26.4	33.4	25.7	32.3	25.1	31.4	4.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.3	19.8	30.6	23.7	19.1	30.1	22.9	18.1	29.2	83.7	33.5	80.5	32.4	77.4	31.3	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	8.6	7.9	-17.9	38.6	3.0	1.8	-20.0	39.9	-21.7	41.0	-23.4	42.0	-25.6	43.3		
(5)			-18.5	27.5	2.8	0.9	-20.5	28.2	-22.1	28.7	-23.7	29.2	-25.7	29.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.1	-0.2	2.2	8.1	13.9	19.4	24.7	27.2	26.3	21.9	15.0	8.0	1.9		
	DBStd	10.82	6.47	6.64	6.41	5.25	4.47	3.39	3.20	3.26	4.33	5.16	5.54	5.91		
	HDD10.0	1054	320	229	112	22	1	0	0	0	14	100	256			
	HDD18.3	2506	575	451	321	149	43	2	0	1	129	311	508			
	CDD10.0	2551	3	12	53	139	293	440	532	506	357	168	41	6		
	CDD18.3	959	0	0	3	17	76	192	274	248	122	25	1	0		
	CDH23.3	10044	0	2	29	140	638	1964	3249	2694	1127	195	6	0		
	CDH26.7	4324	0	0	3	24	180	802	1584	1259	424	48	0	0		
(14) Wind	WSAvg	3.7	3.8	3.9	4.2	4.4	3.8	3.8	3.4	3.2	3.2	3.5	3.7	3.6		
(15) Precipitation	PrecAvg	1021	30	45	61	102	140	152	114	105	108	83	54	40		
	PrecMax	1322	78	111	127	189	288	283	317	255	254	192	172	105		
	PrecMin	713	4	0	20	17	38	47	4	28	19	15	3	0		
	PrecStd	172	18	27	31	41	70	64	77	60	65	46	41	27		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.7	22.4	26.4	29.3	32.4	35.8	38.6	38.4	35.3	30.6	24.0	19.1		
		MCWB	12.5	13.0	16.9	18.7	22.7	24.2	24.4	24.0	23.5	22.0	16.4	14.9		
	2%	DB	15.4	18.6	23.6	26.9	30.4	33.8	36.5	36.0	32.7	27.8	21.6	16.1		
		MCWB	9.9	12.3	15.6	17.7	21.5	24.2	24.9	24.5	23.3	19.7	14.9	12.0		
	5%	DB	12.1	15.5	21.2	24.8	28.7	32.5	34.8	34.1	31.1	25.4	18.9	13.1		
		MCWB	7.3	9.8	14.2	16.6	20.7	23.9	24.9	24.2	22.7	18.4	13.4	8.5		
	10%	DB	8.8	12.5	18.5	22.8	27.1	31.1	33.1	32.6	29.0	23.1	16.7	10.7		
		MCWB	5.0	7.7	12.4	15.8	20.1	23.3	24.4	23.9	21.7	17.5	12.1	6.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.4	15.7	18.7	20.8	24.5	26.4	27.7	27.1	25.6	22.9	18.4	16.0		
		MCDB	17.4	19.8	24.4	26.7	30.2	32.8	34.1	33.8	31.8	28.6	21.8	18.6		
	2%	WB	10.1	12.8	16.9	19.5	23.1	25.5	26.7	26.1	24.3	21.6	16.0	12.6		
		MCDB	14.0	17.2	21.9	24.6	28.7	32.0	33.7	32.9	30.6	26.0	19.9	15.3		
	5%	WB	7.5	10.2	15.3	18.1	21.9	24.7	26.0	25.3	23.5	19.9	14.5	9.1		
		MCDB	11.7	15.2	20.1	23.2	27.3	30.9	32.8	32.0	29.6	24.1	17.9	12.4		
	10%	WB	5.1	7.6	12.8	16.5	20.7	23.9	25.3	24.6	22.6	18.1	12.4	6.8		
		MCDB	8.8	12.3	17.9	22.0	25.8	29.7	31.7	30.9	28.2	22.1	16.3	10.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	9.7	10.9	10.6	9.9	9.5	9.4	9.7	10.2	10.3	9.7	8.9		
		MCDBR	14.0	14.0	14.3	13.1	11.3	10.4	10.5	11.4	11.1	12.3	12.8	12.9		
	5% WB	MCWBR	9.4	9.1	8.1	7.3	5.5	4.5	4.0	4.3	4.7	6.3	8.1	9.0		
		MCWBR	13.1	12.9	12.3	11.3	10.0	9.6	9.4	10.1	10.0	9.9	11.0	11.5		
(40) Clear-Sky Solar Irradiance	taub		0.296	0.308	0.339	0.391	0.431	0.434	0.442	0.441	0.391	0.347	0.305	0.291		
	taud		2.464	2.436	2.408	2.281	2.227	2.254	2.250	2.258	2.352	2.464	2.523	2.502		
	Ebn at Noon		882	919	919	885	852	847	837	828	852	861	866	862		
	Edh at Noon		79	94	107	130	140	136	136	131	112	91	75	71		
(44) All-Sky Solar Radiation	RadAvg		2.19	2.97	3.86	4.93	5.69	6.44	6.42	5.68	4.78	3.39	2.39	1.90		
	RadStd		0.22	0.32	0.34	0.31	0.50	0.49	0.41	0.29	0.32	0.39	0.23	0.18		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (26 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page